

Video Games Sales Using Interactive Power BI Dashboard

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Abstract

The video game industry has experienced significant growth over the past decades, becoming one of the largest entertainment sectors worldwide. This study presents a comprehensive analysis of video game sales using an interactive Power BI dashboard. The dataset includes variables such as game title, platform, genre, publisher, release year, and regional sales (North America, Europe, Japan, and others).

The dashboard provides insights through multiple visualizations, including genre-based analysis, platform comparisons, publisher performance, and regional sales distribution. Key findings indicate that action and sports genres dominate global sales, while North America contributes the highest share of revenue. The analysis also highlights the influence of platforms and publishers on sales performance.

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Keywords

Video Game Sales, Power BI, Data Visualization, Gaming Industry, Sales Analysis

I. Introduction

The video game industry has evolved rapidly, generating billions in revenue annually. Understanding sales trends across different platforms, genres, and regions is crucial for developers, publishers, and marketers.

This project aims to analyze video game sales data using an interactive Power BI dashboard. It helps

in identifying key trends, top-performing genres, and regional preferences, enabling better business decisions in the gaming industry.

II. Related Works

Previous studies have analyzed video game sales using statistical methods and predictive modeling techniques. Researchers have explored how factors such as genre, platform, and publisher impact sales performance.

Data visualization tools have been widely used to present gaming analytics in a user-friendly way. However, many studies focus on prediction rather than interactive exploration.

This project focuses on creating a visual analytics dashboard that allows users to explore sales patterns dynamically across multiple dimensions.

III. DATASET AND PREPROCESSING

A. Dataset Description

The dataset includes the following attributes:

- Game Title
- Platform
- Genre
- Publisher
- Release Year
- North America Sales
- Europe Sales
- Japan Sales
- Other Region Sales
- Global Sales

Donut Chart (Regional Sales Distribution)

Line Chart (Sales Trend by Year)

Tree Map (Publisher-wise Sales Contribution)

B. Data Preprocessing

The following preprocessing steps were performed:

- Removal of missing and duplicate values
- Data type conversion
- Standardization of categorical values (genre, platform)
- Aggregation of regional sales into global metrics

C. Feature Engineering

New insights were derived, including:

- Total Global Sales
- Genre-wise Sales Distribution
- Platform Performance Metrics
- Regional Contribution Analysis

IV. Methodology

A. Dashboard Design

An interactive Power BI dashboard was developed using:

KPI Cards (Total Sales, Top Genre, Top Platform)

Bar Charts (Sales by Genre and Platform)

B. Analytical Approach

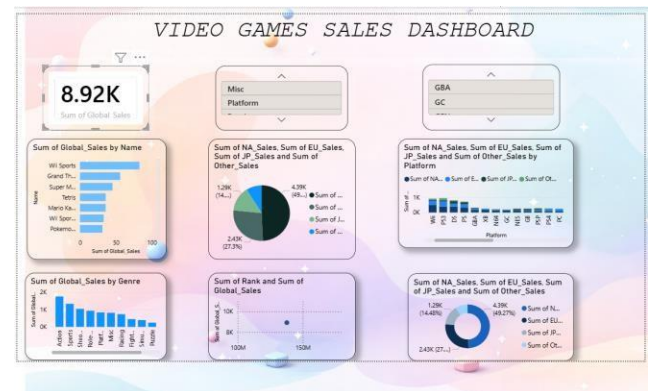
The dashboard enables:

Comparative Analysis

Trend Analysis

Distribution Analysis

Regional Insights



V. Results

The analysis reveals that global video game sales are dominated by specific genres and platforms. Action and sports games generate the highest revenue, while certain platforms consistently outperform others.

A. Genre-Based Analysis

Fig. 2 shows that action and sports genres contribute the highest share of global sales, indicating strong user preference for these categories.

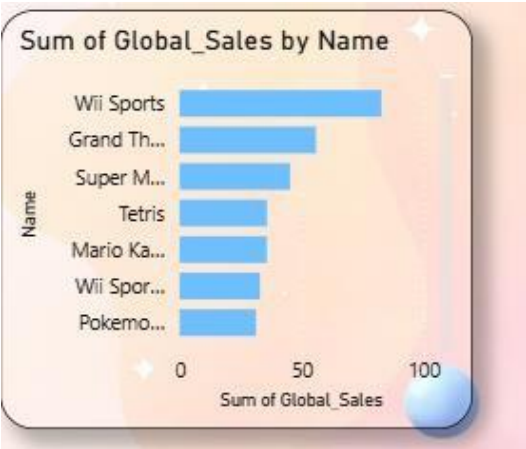


Fig2. This bar chart shows global sales distribution across different genres.

B. Platform-Based Analysis

Fig. 3 highlights sales distribution across different platforms. Consoles such as PlayStation and Xbox show higher sales compared to others.

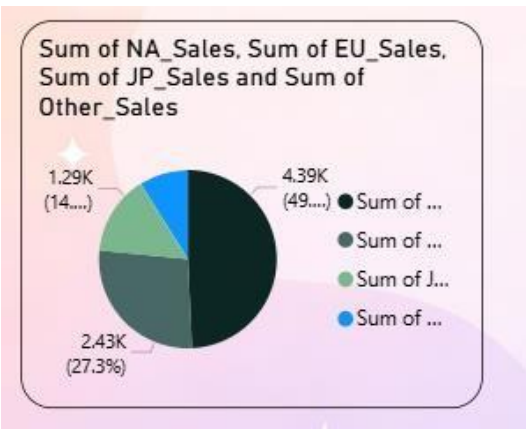


Fig3. This pie chart illustrates regional sales distribution, with North America contributing the largest share followed by Europe.

C. Regional Sales Distribution

Fig. 4 illustrates that North America leads in video game sales, followed by Europe and Japan. This indicates regional differences in gaming preferences.

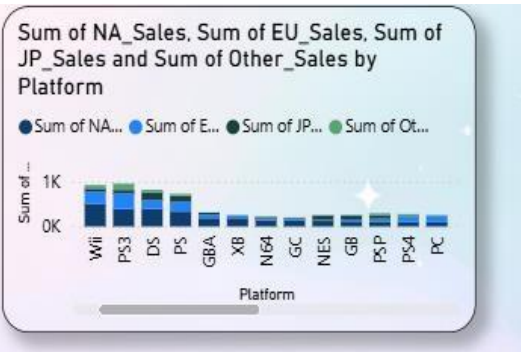


Fig. 4 This chart shows regional sales distribution across platforms, with Wii and PS3 generating the highest overall sales.



Fig. 5 This scatter plot shows the relationship between game ranking and global sales, indicating how higher sales correspond to better ranks.

E. Publisher Analysis

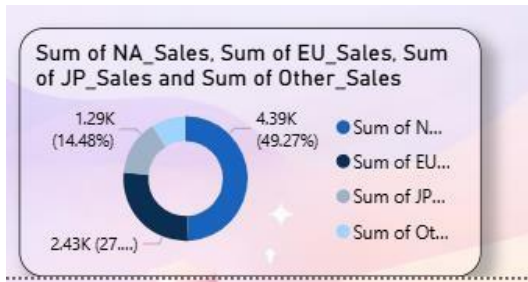


Fig. This donut chart shows regional sales distribution, with North America leading, followed by Europe, while Japan and other regions contribute smaller shares.

F. Overall Dashboard Insights

- **Fig. 1:** KPI overview of total sales and top categories
- **Fig. 7:** Full dashboard view without filters
- **Fig. 8:** Dashboard filtered by genre
- **Fig. 9:** Dashboard filtered by platform
- **Fig. 10:** Comparative insights using multiple filters

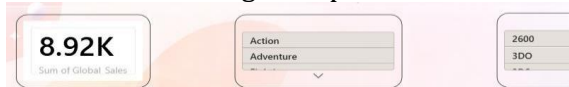


Fig. 7

This section highlights the total global sales (8.92K) along with filters for genre and platform selection.

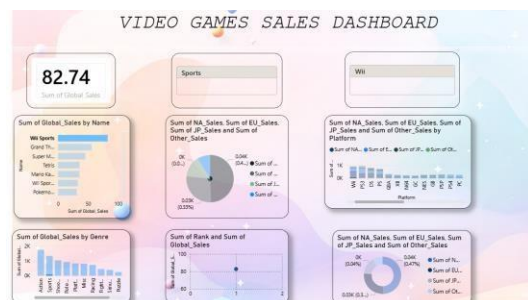


Fig. 8. Overall Dashboard View

This dashboard provides a comprehensive overview of video game sales, including global metrics, regional

distribution, platform performance, and genre-based insights.

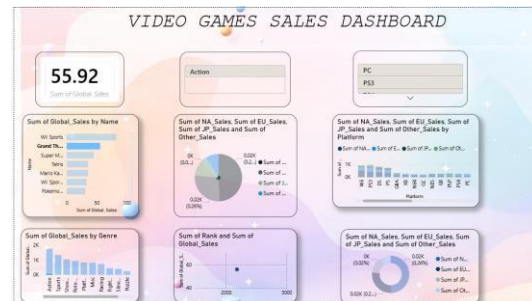


Fig. 9. This dashboard presents video game sales insights with updated filters, showing global sales, regional distribution, and performance across genres and platforms.

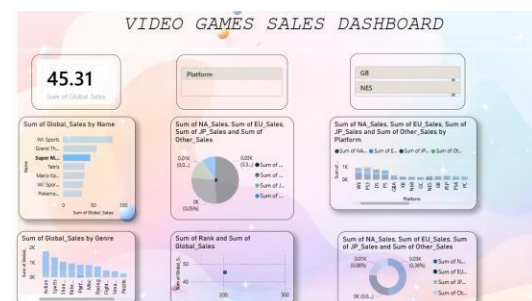


Fig. 10. This dashboard shows filtered video game sales insights by platform, highlighting global sales, regional distribution, and performance trends.



This dashboard displays video game sales

insights filtered by puzzle genre and platform, highlighting global sales and regional trends.

VI. Conclusion

This project presents an interactive Power BI dashboard for analyzing video game sales data. The findings show that genre, platform, and region significantly influence sales performance.

Action and sports genres dominate the market, while North America remains the leading region in terms of revenue. The dashboard provides clear and interactive insights, making it easier to understand complex sales patterns.

Future work can include predictive models to forecast sales trends and analyze player behavior for better decision-making

VII. REFERENCES

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